

Project Report

A Study Of Sandha Yantra : When Shadows Become Science – Traditional Indian Astronomical Instrument and Its Digital Simulation.

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1. Abstract

The Sandha Yantra (also known as Shanku Yantra) is a traditional Indian astronomical instrument based on the observation of the shadow cast by a vertical rod. It was used to study the position of the Sun and obtain information related to direction, solar time and the Sun's altitude. The instrument demonstrates the practical application of geometry, trigonometry and astronomical observation.

As part of this project, a digital simulation of the Sandha Yantra was developed to demonstrate its working principle. The simulation represents the Sandha, the Sun, the ground surface and the resulting shadow. By changing the position of the Sun, the simulation demonstrates how the length and direction of the shadow change. This provides a visual and interactive way to understand the relationship between the Sun's position, shadow length and solar altitude.

2. Introduction

The Sandha Yantra is one of the traditional astronomical instruments associated with the scientific and mathematical practices of ancient India. **Sandha** (derived from the Sanskrit root word **sandhi**) literally means "union" or "intersection,". In the context of astronomical or geometric instruments (*yantras*), the term is used to describe a mechanism or design built upon an intersection or coupling of parts. The basic principle of the instrument is the

observation and measurement of the shadow produced by the vertical rod when exposed to sunlight.

Before the development of modern astronomical equipment, observations of the Sun and other celestial bodies were carried out using carefully designed instruments and mathematical methods. The Sandha Yantra provided a relatively simple method of studying the apparent movement of the Sun.

The length and direction of the shadow change throughout the day as the apparent position of the Sun changes. By measuring these changes, various astronomical and geographical observations can be made.

The Sandha Yantra is particularly interesting from an educational perspective because its operation combines **astronomy, geometry, trigonometry and measurement**. It also demonstrates how a simple physical arrangement can be used to study a natural phenomenon quantitatively.

3. Construction of Sandha yantra

The basic Sandha Yantra consists of a vertical rod or pole, known as the Shanku, placed on a horizontal and preferably level surface. When sunlight falls on the rod, it produces a shadow on the surface.

The length and direction of this shadow depend on the position of the Sun. During the morning, when the Sun is relatively low in the sky, the shadow is comparatively long. As the Sun rises higher, the shadow becomes shorter. Around local solar noon, the shadow reaches its minimum length. It then increases again as the Sun moves towards the western part of the sky.

The direction of the shadow also changes during the day. These changes can be observed and recorded to study the apparent motion of the Sun.

4. Advantages of Sandha Yantra

1. It demonstrates a direct relationship between observation and mathematical calculation.
 2. It can be used to study the apparent movement of the Sun.
 3. It provides a practical application of trigonometry.
 4. It has a simple construction.
 5. Its basic principle can be understood without complex astronomical equipment.
 6. Its working principle can be effectively reproduced using a digital simulation.
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5. Working Principle

The Sandha Yantra operates on a simple geometric and trigonometric principle. The vertical rod, its shadow and the direction of sunlight form a geometric relationship that can be used to determine the Sun's altitude. If the height of the gnomon is known and the length of its shadow is measured, the altitude of the Sun can be calculated using:

$$\tan \theta = \text{Height of Rod} / \text{Length of Shadow}$$

$$\text{therefore, } \theta = \tan^{-1} (\text{Height of Rod} / \text{Length of Shadow})$$

For a fixed Shanku height, a shorter shadow corresponds to a higher apparent position of the Sun, while a longer shadow corresponds to a lower apparent position. The direction of the shadow also changes as the Sun appears to move across the sky. By recording the shadow at different times, it is possible to study the apparent daily movement of the Sun.

6. Applications of Sandha Yantra

6.1 Determination of Direction - The changing direction of the shadow can be used to identify approximate geographical directions. Careful observations can help establish the east-west direction and, consequently, the north-south direction.

6.2 Estimation of Solar Time - The position of the shadow changes with the apparent movement of the Sun. This principle can be used to estimate solar time when appropriate reference markings or calibration are available.

6.3 Measurement of Solar Altitude - The height of the Rod and the length of its shadow can be used with trigonometry to calculate the altitude of the Sun.

6.4 Astronomical Observation - The instrument provides a simple method of observing the apparent motion of the Sun and recording changes in its position during the day.

6.5 Study of Seasonal Changes - Observations made at different times of the year can demonstrate changes in the Sun's apparent path and altitude, helping in the study of seasonal variations.

7. Digital Simulation of Sandha Yantra

As part of this project, a **digital simulation of the Sandha Yantra** was developed to demonstrate the working principle of the instrument in a virtual environment. The major components of the Sandha Yantra include:

1. A vertical rod
2. The ground surface
3. The Sun
4. The shadow produced by rod
5. the changing position of sun

The main function of the simulation is to visually demonstrate how the **position of the Sun affects the length and direction of the shadow**. When the position or angle of the Sun is changed in the simulation, the corresponding shadow changes automatically. This allows the user to observe the same basic phenomenon that would occur with a physical Sandha Yantra.

The simulation also illustrates the mathematical relationship between the height of the rod, shadow length and solar altitude. For example, if the height of the rod remains constant while the Sun moves higher in the simulated sky, the shadow becomes shorter. Conversely, when the simulated Sun is lower, the shadow becomes longer.



The simulation therefore serves as an interactive learning tool. Instead of only reading about the working principle, the user can observe the relationship between the variables directly.

8. REFERENCES

9. General references on traditional Indian astronomy and astronomical instruments.
10. References on gnomons, sundials and shadow-based astronomical observations.
11. Standard mathematics references for trigonometric relationships.
12. Project simulation and observations developed as part of this study.

9. Team Contribution

10. Ahan Kulawade & Arjun Nigade - **Simulation Development and Technical Implementation**
11. Anand Tiwary & Sumedha Adake - **Project Presentation and Oral Communication**
12. Hemal Narkhede & Tanvi Patil - **Technical Documentation and Report Preparation**
13. Tanvi Patil & Hemal Narkhede - **Report Compilation and Final Documentation**

Signature :
